

**INTERNATIONAL INSTITUTE OF INFORMATION
TECHNOLOGY, HYDERABAD****CE8.401: Disaster Management****In-Class Quiz***Spring 2026***Student Name:** _____ **Roll Number:** _____**Section A: Single Choice MCQs (20 Questions)***Select the most appropriate option for each question.*

1. Which of the following best describes the "Mitigation" phase in the disaster management cycle?
 - (a) Immediate actions taken during a disaster to save lives.
 - (b) Rebuilding infrastructure to standard specifications after a disaster.
 - (c) Efforts to reduce loss of life and property by lessening the impact of disasters.
 - (d) Planning alternate routes and stockpiling emergency food supplies.
2. According to the 2018 Kerala Floods case study, what was the primary conclusion drawn from the HEC-HMS hydrological modeling of the Periyar River Basin?
 - (a) Proper reservoir operations could have attenuated the peak flood by over 80%.
 - (b) The disaster was solely caused by the structural failure of the Mullaperiyar dam.
 - (c) Reservoir operations could only provide a 16–21% peak attenuation even if emptied in advance.
 - (d) The intermediate catchments without reservoirs contributed no significant runoff to the flooding.
3. In the context of risk assessment, which type of uncertainty is associated with the natural variation of the input data (e.g., variations in soil characteristics)?
 - (a) Epistemic uncertainty
 - (b) Aleatory uncertainty
 - (c) Ontological uncertainty
 - (d) Deterministic uncertainty
4. Which organization was restructured from the Enforcement Vigilance and Disaster Management Department (EV&DM) to safeguard public properties and coordinate with NDRF in Hyderabad?
 - (a) GHMC-DRF
 - (b) HYDRAA
 - (c) TS-SDMA

- (d) NDMA-South
5. Based on the IMD classification for drought, a region experiencing a rainfall deficiency of 35% compared to normal rainfall is facing a:
- (a) Mild Drought
 - (b) Moderate Drought
 - (c) Severe Drought
 - (d) Extreme Drought
6. Which satellite-based indicator is used to detect groundwater depletion by measuring anomalies in the Earth's gravitational field?
- (a) NDVI
 - (b) LST
 - (c) CHIRPS
 - (d) GRACE
7. In Quantitative Risk Assessment (QRA), what does the expression $P(S|H)$ specifically quantify?
- (a) Temporal probability of a certain hazard scenario occurring.
 - (b) Spatial probability that a particular location is affected given a certain hazard.
 - (c) Vulnerability of the elements at risk given the hazard intensity.
 - (d) Quantification of exposed elements-at-risk for a given hazard.
8. A situation where receded ground water levels fall below 15m of depth, rendering it unavailable for common use, is defined as which type of drought?
- (a) Meteorological Drought
 - (b) Agricultural Drought
 - (c) Hydrological Drought
 - (d) Socioeconomic Drought
9. In cloud seeding, which of the following compounds is typically dispersed as an aerosol to act as ice nuclei?
- (a) Sodium chloride
 - (b) Calcium chloride
 - (c) Silver iodide
 - (d) Potassium nitrate
10. According to the "swing analogy" for building dynamics, mass unsymmetry in a building undergoing earthquake ground shaking primarily causes:
- (a) Pure lateral displacement
 - (b) Twisting (torsion) about a vertical axis
 - (c) Vertical settlement
 - (d) Resonance with long-period waves only

11. Which IS code provides the criteria for the earthquake-resistant design of structures in India?
 - (a) IS 456
 - (b) IS 13920
 - (c) IS 1893
 - (d) IS 875
12. The CAG report on the 2018 Kerala Floods highlighted several institutional failures. Which of the following was explicitly mentioned as a missing preparedness component in the state?
 - (a) Lack of structural dams
 - (b) Absence of flood forecasting stations
 - (c) No SDRF battalions deployed
 - (d) Complete absence of the NDRF
13. In epidemiological risk analysis, if A represents the number of incidence cases in an exposed group, and B represents the non-incidence cases in that same group, how is the Incidence Risk among the exposed group calculated?
 - (a) A/B
 - (b) $A/(A + B)$
 - (c) $(A + B)/A$
 - (d) $B/(A + B)$
14. Which Global Risk Indicator measures a country's risk from a macroeconomic and financial perspective regarding possible catastrophic events?
 - (a) Local Disaster Index (LDI)
 - (b) Prevalent Vulnerability Index (PVI)
 - (c) Disaster Deficit Index (DDI)
 - (d) Risk Management Index (RMI)
15. According to FEMA's best practices for flood mitigation, which of the following is considered "Hard Engineering"?
 - (a) Diverting flood water to natural reservoirs
 - (b) Wetland restoration
 - (c) Stone-fills and lining of canals
 - (d) Land-use zoning
16. Which on-farm water management strategy involves covering the soil with plant material to reduce soil moisture evaporation?
 - (a) Conservation tillage
 - (b) Mulching
 - (c) Cover cropping
 - (d) Furrow irrigation

17. In the context of building dynamics, what is the approximate fundamental natural period (T) of a 15-storey building?
- (a) 0.05 sec
 - (b) 0.4 sec
 - (c) 1.0 sec
 - (d) 4.0 sec
18. When analyzing risk curves (Loss vs. Temporal Probability), uncertainty in the hazard intensity and vulnerability is visualized by:
- (a) A single deterministic data point.
 - (b) A perfectly linear graph.
 - (c) Rectangles defined by the variation in probability and losses.
 - (d) A single step-function curve.
19. Which tool is used in risk assessment to graphically analyze the sequence of subsequent events branching out from a single initial event?
- (a) Risk Matrix
 - (b) Indicator-Based Assessment
 - (c) Fault-Tree Analysis
 - (d) Event-Tree Analysis (EVA)
20. What is the mathematical relationship between Duty (D), Base period (B in days), and Delta (Δ in meters) in irrigation management?
- (a) $D = \frac{8.64\Delta}{B}$
 - (b) $D = \frac{8.64B}{\Delta}$
 - (c) $\Delta = \frac{8.64D}{B}$
 - (d) $B = \frac{8.64D}{\Delta}$

Section B: Fill in the Blanks (10 Questions)

Write the correct word or phrase in the space provided.

21. The total amount of water depth required by various crops for full growth during their base period is known as _____.
22. _____ drought occurs when a water shortage significantly damages or destroys agricultural crops.
23. In Event-Tree Analysis, the risk levels of specific outcomes are calculated by _____ the probabilities of the branches along the path to that outcome.
24. IS _____ provides the code of practice for ductile detailing of reinforced concrete structures subjected to seismic forces.
25. _____ uncertainty is associated with incomplete or imperfect knowledge about the processes involved and a lack of sufficient data.

26. The _____ Index brings together a group of indicators that measure a country's organizational and institutional performance in reducing vulnerability and recovering from disasters.
27. _____ cloud seeding disperses salts like sodium chloride or calcium chloride to absorb water vapor from the atmosphere.
28. The "virgin simulation" of a hydrological model analyzes the stream flow assuming that dams and reservoirs are _____.
29. In terms of absolute economic risk, the metric generated by averaging potential financial damages over a given time horizon is called the Average Annual _____.
30. A _____ matrix is a qualitative risk mapping approach used to evaluate risks by plotting likelihood on one axis and impact on the other.

Section C: True or False (5 Questions)

State whether the following statements are True (T) or False (F).

31. _____ The 2018 Kerala Floods were entirely preventable if the reservoir operations at the 17 dams in the Periyar basin had been managed perfectly.
32. _____ The fundamental natural period (T) of a standard elevated water tank is typically much longer than that of a single-storey building.
33. _____ In risk assessment, if the Relative Risk calculation yields a value larger than 1, it implies that exposure to the factor has increased the incidence of risk.
34. _____ Sprinkler irrigation is highly precise and generally considered more efficient with less evaporative wastage compared to drip irrigation.
35. _____ The Local Disaster Index (LDI) is primarily utilized to measure the financial and macroeconomic deficit a country faces post-disaster.

Section D: Matching (5 Questions)

Match the items in Column A with their most appropriate corresponding items in Column B. Write the correct number from Column B next to the letter in Column A.

Table 1: Satellite Indicators and Their Applications

Column A	Column B
36. NDVI _____	1. Groundwater anomaly via gravity detection
37. LST _____	2. Rainfall estimates
38. GRACE _____	3. Vegetation health and crop stress
39. CHIRPS _____	4. Land surface temperature
40. IS 875 _____	5. Code of practice for design loads